

PRAIGUKTI

AI-Powered Project Intelligence for Indian Public Infrastructure Monitoring

A decision-support platform that predicts infrastructure project delays before they happen, explains *why*, proves it with historical precedent, screens the site against protected geography, and recommends the exact interventions to take first.

Presentation Content Pack · Slide-by-slide script, talking points and roadmap

How to use this document. Each numbered block below is one slide. It gives you: the slide title, the bullets to put on the slide (keep them short on screen), a "Say this" speaker script, and the supporting detail you may be asked about in Q&A. Status tags used throughout — **WORKING** built and demonstrable today, **PARTIAL** built but running in a limited/simulated mode, **ROADMAP** designed with the code seam in place, to be completed next. Section 14 lists every non-working item in one place, honestly, with the plan for each.

0 Deck Structure — 18 slides

1. Title
2. The problem, in numbers
3. Why existing monitoring fails
4. Our solution in one line
5. System overview / architecture
6. Feature 1 — AI delay risk prediction
7. Feature 2 — Explainability (SHAP)
8. Feature 3 — Historical similarity engine
9. Feature 4 — Priority & decision engine
10. Feature 5 — GIS boundary screening
11. Feature 6 — Spatial analysis engine
12. Feature 7 — Intervention recommendations
13. Feature 8 — Document intelligence
14. Feature 9 — SLA rule monitoring & alerts
15. Feature 10 — Portfolio dashboard & scenario lab
16. The AI/ML stack and why each model was chosen
17. What works today vs. what comes next
18. Impact, feasibility and closing ask

1 Slide 1 — Title

SLIDE 1

PRAGUKTI — AI-Powered Project Intelligence for Indian Infrastructure

On the slide:

- Product name: **PRAGUKTI** (from Sanskrit *prāgukti* — "foretelling / prior statement")
- Expansion: **P**roject **A**nalytics & **I**ntelligence for **M**onitoring **A**ctions with **N**ational **A**ccountability
- Tagline: *"Every project. One view. Acted on before it slips."*
- Problem Statement SIH26103 · Team name · Institution

"Indian infrastructure does not usually fail because nobody was watching. It fails because by the time the monthly report reaches the desk that can act, the decision window has already closed. PRAGUKTI is built to close that gap."

2 Slide 2 — The Problem, In Numbers

SLIDE 2

Delay and cost overrun are the default, not the exception

On the slide (4 stat tiles + one line):

~40%

OF MAJOR PROJECTS
RUN LATE

₹ '000s Cr

LOCKED IN COST
ESCALATION

Months

LOST BEFORE A
DELAY IS EVEN
VISIBLE

Reactive

CURRENT
MONITORING
POSTURE

Talking points:

- Central-sector infrastructure projects are monitored by **periodic, self-reported progress statements**. A problem becomes visible only after it has already cost time.
- The four causes that repeat, project after project: **land acquisition not resolved, statutory/environmental clearance pending, funding release delays, contractor non-performance** — plus milestone slippage that compounds all four.
- Monitoring today answers *"what is the status?"*. Nobody is answering *"which 30 of my 5,000 projects will slip next quarter, and what do I do about them this week?"*
- There is **no institutional memory**: the same failure repeats in a new state because nobody surfaces *"this has happened 47 times before, and here is what worked."*

"The data to predict these failures already exists inside the system. It is just never used predictively. That is the entire gap we are attacking."

3 Slide 3 — Why Existing Monitoring Fails

SLIDE 3**Four structural gaps in current project monitoring**

Gap	What happens today	What it costs
Reactive, not predictive	Status is recorded after the fact; dashboards report the past	Intervention arrives after the decision window has closed
No prioritisation	Thousands of projects, all shown equally; attention goes to whoever escalates loudest	Genuinely critical projects compete with noise
No institutional memory	Each project is judged in isolation; past comparable projects are not consulted	Known, solved failure modes are re-learned at full cost
Siloed geography	Environmental/forest conflicts are discovered during clearance, not at siting	Re-alignment after commitment — the most expensive possible moment

"Every one of these four gaps maps to one engine in our system. We did not build features and look for problems — we took the four failure modes and built one engine each."

4 Slide 4 — Our Solution In One Line

SLIDE 4

From "what is the status" to "what do I do on Monday"

On the slide — the one-line promise:

PRAGUKTI converts raw project records into a ranked, explained, evidence-backed action list — so the scarcest resource in governance, senior attention, goes to the right project at the right time.

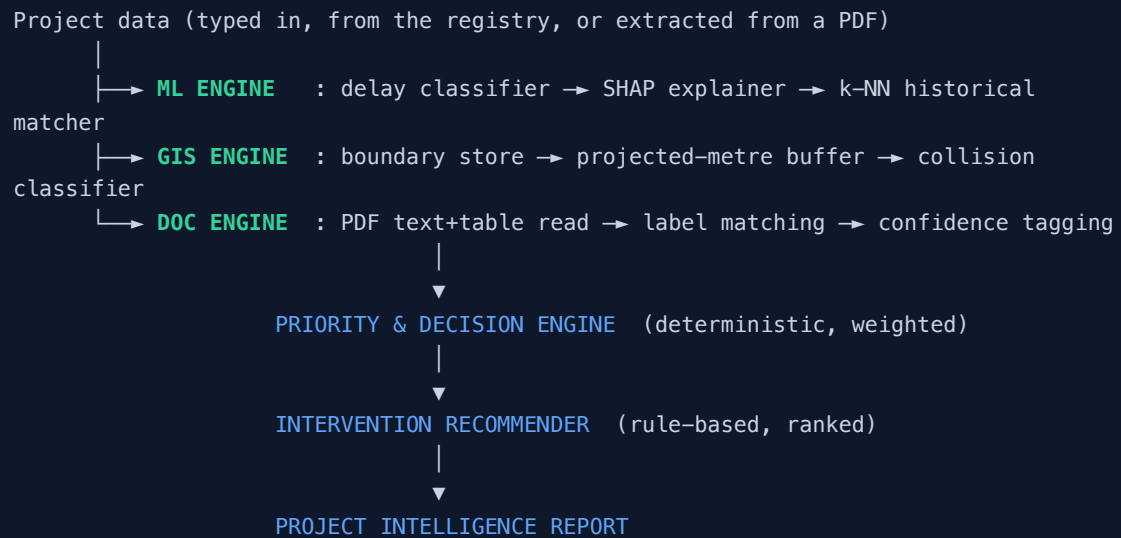
The five questions the system answers for every project:

1. **Will it be delayed?** → calibrated probability from a trained classifier
2. **Why?** → the top five contributing factors, ranked, in plain language
3. **Has this happened before?** → the most similar past projects, their real outcomes, and what interventions actually worked on them
4. **Where does it sit against everything else?** → a 0–100 composite priority score
5. **What should be done first?** → a ranked intervention list with statutory obligations above managerial ones

"Notice that none of those five answers is 'here is a number, trust us'. Every single output carries its reason with it. In government, an unexplainable recommendation is an unusable recommendation."

5 Slide 5 — System Overview

One pipeline, three intelligence engines, one report



Design principles to state out loud (these win marks):

Principle	What it means in this system
Deterministic & auditable	Identical inputs always produce identical outputs. Every threshold and weight lives in a named config file, not scattered as magic numbers. No randomness at inference time.
Explainability first	SHAP for the ML prediction, template-based rationale for every intervention, geometry-derived reasons for every GIS flag.
Measured vs. predicted, never blurred	Portfolio figures and SLA breaches are <i>facts computed from the record</i> . Model outputs are labelled as estimates, with confidence. The UI never mixes the two.
Graceful degradation	If the ML, GIS or similarity service fails to load, the application still starts and reports exactly which capability is unavailable — it never silently substitutes a fake value.
No silent repairs	Invalid geometry, malformed data and unreadable PDFs are rejected with a reason, never auto-"fixed" into something plausible-looking.

A trained classifier that answers "will this project slip?"

Raw project record → Feature engineering → XGBoost predict_proba → calibrated delay probability (0-1) → risk band

Aspect	Detail
Model	XGBoost gradient-boosted decision trees, binary:logistic objective
Why this model	Tabular, mixed numeric + categorical, non-linear interactions, small-to-medium data — the regime where gradient boosting reliably beats both linear models and neural networks. It also gives exact tree-based SHAP attribution, which a neural net cannot.
Input signals	Land acquisition pending, pending clearances, funding issue, contractor issue, milestones total vs. delayed, physical progress, financial progress, original vs. revised cost, planned duration, project age, previous schedule deviation, sector, state
Engineered features	cost_overrun_percentage, financial_physical_gap, milestone_delay_ratio, project_age_ratio — derived ratios that carry far more signal than the raw fields alone
Preprocessing	Median imputation (numeric) + most-frequent imputation and one-hot encoding (categorical), all inside one scikit-learn Pipeline so training and inference can never diverge
Training data	5,000 project records across 8 sectors and 15 states, with labelled delay outcomes (≈2,000 delayed / 3,000 on-track — a realistic, non-degenerate class balance)
Train/test split	4,000 train / 1,000 held-out test, stratified
Measured performance	Accuracy 0.697 · Precision 0.650 · Recall 0.535 · F1 0.587 · ROC-AUC 0.753
Serving	Pipeline trained offline and persisted with joblib; loaded once at API startup. Inference is a single forward pass — milliseconds, no retraining on the request path.

Be honest about the metrics — it is a strength, not a weakness. ROC-AUC of 0.75 on a genuinely hard, multi-factor social-system prediction is a useful signal, not a solved problem. Say this: *"We report the real held-out numbers rather than a cherry-picked accuracy. 0.75 AUC means the model reliably ranks a delayed project above an on-track one three times out of four — which is exactly what a triage tool needs to do. And because the model is deliberately shallow and regularised, it does not pretend to more certainty than the data supports."*

"The output is never a yes/no verdict. It is a probability with an explanation attached, which an officer can agree or disagree with — and the disagreement is itself informative."

SLIDE 7

Every prediction arrives with its reasons

Aspect	Detail
Technique	SHAP (SHapley Additive exPlanations), TreeExplainer — the exact, game-theoretic attribution for tree ensembles, not an approximation
What it produces	The top 5 factors driving <i>this specific</i> prediction, each with a signed contribution and a normalised importance share
Human translation layer	One-hot encoded feature names are folded back to their logical field (e.g. every <code>state_*</code> column recombines into "State") and mapped to plain-language labels: "Land acquisition", "Pending clearances", "Milestone delay ratio", "Financial-physical progress gap", "Previous schedule deviation"
Performance design	The explainer is constructed once when the model is loaded, not once per request
Why it matters	A government officer cannot act on "risk = 0.81". They can act on "81% — driven primarily by land acquisition pending (34% of the signal) and 7 of 12 milestones already slipped (29%)"

Slide visual suggestion: a horizontal bar chart of the five factors with their percentage contributions, red for risk-increasing and green for risk-reducing. This is the single most persuasive visual in the deck — it is the moment the judges see the black box open.

"This is the difference between an AI that gives orders and an AI that gives evidence. Public administration can only use the second kind."

SLIDE 8

Institutional memory, made queryable

Project features → engineering → `StandardScaler` + `OneHotEncoder` → domain feature weighting → k-NN search → top-k matches + aggregated evidence

Aspect	Detail
Algorithm	k-Nearest Neighbours, Euclidean distance over a scaled and encoded feature space
Why k-NN and not a deep model	The output must be <i>the actual past projects themselves</i> , with names and real recorded outcomes. A neighbour search returns evidence an officer can independently verify; an embedding model would return a number they cannot check.
Corpus	1,500 completed historical projects, each carrying its real outcome: actual delay in months, actual cost overrun %, final status, primary delay cause, intervention taken, and whether that intervention worked
Domain feature weighting	Not all fields are equally predictive, so distance is weighted: milestone_delay_ratio 2.5× , land_acquisition_pending 2.0× , funding_issue 2.0× , previous_schedule_deviation 1.75× , financial_physical_gap 1.75× , everything else 1.0×. These are a documented priority ordering validated against the historical outcomes — stated as such, not passed off as a fitted parameter.
Aggregated evidence returned	% of similar projects that slipped more than 6 months, average delay severity, average cost overrun, the dominant primary delay cause, and the interventions that succeeded on those matches
Recorded delay causes in the corpus	Land acquisition (313), milestone slippage (98), other execution delay (77), pending clearance (34), funding release (22), contractor performance (16), no material delay (940)

"When the system says 'act on land acquisition first', it is not an opinion. It is: here are 40 projects that looked exactly like yours, 31 of them slipped, land acquisition was the cause in most, and here is the intervention that pulled the rest back."

One 0–100 score that makes thousands of projects triageable

Deliberately NOT machine learning. This layer is a transparent weighted composite, because the ranking that decides where ministerial attention goes must be defensible line by line and tunable by policy — not learned from data nobody can audit.

Component	Weight	Evidence source
AI delay risk	0.35	XGBoost probability (Feature 1)
Predicted delay severity	0.20	$P(\text{delay}) \times \text{average actual delay of similar historical projects}$ — an expected-value estimate of months likely lost
Financial exposure	0.20	Revised cost $\times$ projected overrun trend, normalised on a log scale so one very large project cannot dominate the scale for every smaller one
Historical evidence	0.15	60% "how often comparable projects failed" + 40% "by how much" (Feature 3)
Urgency / schedule position	0.10	Project age vs. planned duration, and proximity to planned completion

The GIS-aware profile — a design detail worth calling out. When a project has coordinates, a second weight profile activates: GIS spatial risk enters at **0.10**, funded by trimming risk $0.35 \rightarrow 0.30$ and urgency $0.10 \rightarrow 0.05$. A project assessed *without* coordinates scores exactly what it scored before the GIS module existed — so adding a new signal never silently re-scores historical assessments. This is backward-compatibility as an explicit engineering commitment.

Output: composite 0–100 → **CRITICAL / HIGH / MEDIUM / LOW**, with the component breakdown always visible so a user can see which of the five drove the band.

"Every weight on this slide lives in one configuration file. A ministry that wants to weight financial exposure higher than schedule can do it in one line, with no code change and no retraining. Policy stays with the policymaker."

10 Slide 10 — Feature 5: GIS Boundary Data Layer

PARTIAL — demo dataset

SLIDE 10

Restricted geography as a first-class, validated data layer

Aspect	Detail
Geometry model	Polygon and MultiPolygon with full RFC 7946 GeoJSON validation
Validation performed	Ring closure, winding-order normalisation, self-intersection rejection, coordinate-bounds checking. Invalid geometry is rejected with a reason , never silently repaired into something that looks valid.
Boundary categories supported	Wildlife sanctuary, national park, tiger reserve, forest reserve, Ramsar wetland, eco-sensitive zone, heritage site, coastal regulation zone, tribal area, defence zone, river/floodplain
Spatial indexing	Bounding-box pre-filter (deliberately over-inclusive, never misses a candidate) → exact geometry intersection test on the survivors
Identity strategy	UUIDv5, derived deterministically from source + name + category, so re-importing the same official dataset produces the same IDs rather than duplicates
Storage	Dual-backend by design: GeoJSON file store (in use) or PostGIS with GiST indexing (production seam, see roadmap)

State this honestly: the boundary store currently holds a **6-boundary demonstration set** — one each of national park, wildlife sanctuary, tiger reserve, forest, Ramsar wetland and eco-sensitive zone — sufficient to prove every code path. The importer, validator and deterministic-ID scheme are built precisely so that ingesting the full national protected-area dataset is a data operation, not a development project. *See Section 14, item 3.*

11 Slide 11 — Feature 6: Spatial Analysis Engine **WORKING**

Geometrically correct proximity analysis — the detail most teams get wrong

Lat/Lon → validate → project WGS84 → AEQD centred on the project point → buffer in metres → spatial index pre-filter → exact intersection → classify → reproject to WGS84 for the map

The one-line technical claim to make on stage: *"Metres are never applied to degrees."* A 5-kilometre buffer computed naively in latitude/longitude is wrong by a different amount in Kerala than in Ladakh, because a degree of longitude is not a fixed distance. We reproject to an Azimuthal Equidistant CRS **centred on the project's own coordinate**, buffer there in true metres, then come back. This is the difference between a screening tool a ministry can rely on and a demo that looks right on a map.

Collision class	Meaning	Action implied
DIRECT COLLISION	The project point lies inside a protected boundary	Siting review — statutory
BUFFER COLLISION	The project's buffer zone intersects a boundary	Clearance review — statutory
PROXIMITY WARNING	A boundary lies within the configured proximity threshold	Flag and monitor
CLEAR	No boundary within the screened radius	No spatial constraint found

Delivered to the user as: an interactive map layer showing the project point, its buffer ring, and every colliding or nearby boundary, each with its category, distance and severity band — plus the same result as structured data that feeds the priority score and the intervention list.

"This screening happens at siting time — which is when a conflict is cheap to fix. Today it is typically discovered during clearance, which is the most expensive possible moment to find it."

SLIDE 12

The system does not stop at a diagnosis

Rule-based and deterministic — no model, no LLM. The same evidence always yields the same ordered list. That reproducibility is what makes a recommendation auditable and, ultimately, defensible in a file note.

Recommendation	Class	Fires when
Initiate environmental clearance review	Statutory	GIS screening returns clearance flags
Commission a siting / alignment review	Statutory	Direct collision with a protected boundary
Accelerate land acquisition resolution	Execution	Land acquisition pending
Escalate regulatory clearance	Statutory	Clearance pending
Milestone recovery plan	Execution	Milestone delay ratio ≥ 0.30
Expedite funding release	Financial	Funding issue flagged
Cost baseline review	Financial	Cost overrun $\geq 15\%$
Contractor performance review	Contractual	Contractor issue flagged
Schedule re-baselining review	Execution	Project past 85% of planned duration with high risk probability

Two design decisions to highlight:

- **Statutory obligations outrank managerial ones** (base weight 100/95 vs. 90 and below) — because no amount of milestone recovery helps a project that cannot lawfully proceed until a clearance is resolved. The ranking encodes that dependency.
- **Capped at 5 recommendations.** Beyond a handful, a "priority list" stops being one. Each carries a rationale sentence populated from the project's own evidence and an expected-impact sentence.

Upload a DPR or progress report — get a filled, reviewable form

PDF upload → text + table extraction → label-based field matching → unit normalisation → confidence tag per field → human confirmation → ML pipeline → saved report

Aspect	Detail
Extraction method	Label-based matching over the text lines and table rows parsed from the PDF. The system looks for the labels it knows; it does not guess or infer a value it cannot find.
Fields recovered	All project risk fields — original and revised cost, planned duration, project age, physical and financial progress, milestone counts, the four issue flags, coordinates
Unit handling	Automatic normalisation across crore / lakh / million / billion, so a report written in lakh and one written in crore produce the same internal value
Confidence	Every extracted field is tagged high or low with an explanatory source note showing where in the document the value came from
Human-in-the-loop	Extracted values are always presented for confirmation before anything reaches the prediction pipeline. Missing fields are shown as empty and must be filled manually — never auto-guessed.
Failure behaviour	A scanned image PDF with no text layer returns an explicit message saying so, rather than silently extracting nothing and appearing to succeed
Repository	Uploaded documents are stored per project with type and description, and confirmed analyses are saved and re-openable from the Analyses page

"We deliberately did not put a language model here. A hallucinated cost figure that silently enters a risk calculation is far worse than an empty field that a human fills in. Extraction is deterministic; the human confirms; only then does the model see it."

14 Slide 14 — Feature 9: SLA Rule Monitoring & Escalation

PARTIAL — alerts previewed, not dispatched

SLIDE 14

Five policy rules that measure breaches as facts, not forecasts

Rule	Computed as	Default threshold
Cost escalation	(revised – original) / original, as %	> 10%
Progress shortfall	Physical progress vs. linear expectation from elapsed time	> 20 points behind
Reporting divergence	physical progress – financial progress	> 15 points
Milestone slippage	milestones delayed / milestones total	> 25%
Schedule overrun	project age – planned duration, in months	> 0 months

Severity: a breach at or beyond 2× its threshold (or beyond 6 months for schedule overrun) is **CRITICAL**; otherwise **WARNING**. Severity is used only for ordering and labelling — never fed into any score.

The key distinction to make on this slide: an SLA breach is a **measured fact about the record** — a specific number crossed a specific published threshold. It is not a prediction and it is not model output. Users can see the rule, the value and the threshold side by side. This is deliberately the most trustworthy surface in the product, and it sits next to the ML predictions without being confused with them.

Escalation: a deadline-tracking SLA engine per project milestone, with an amber threshold at 30 days remaining and a critical threshold at ≥80% risk probability. An SMS escalation path to the ministry monitoring cell is integrated (MSG91 flow API), with recipient masking and a full alert preview. **It is currently in preview-only mode — the alert is composed and shown, never dispatched.** See Section 14, item 2.

15 Slide 15 — Feature 10: Portfolio Dashboard & Workspace **WORKING**

The daily surface: 5,000 projects, one screen

Portfolio overview — every figure computed server-side from the recorded registry, nothing predicted:

- **KPI strip** — projects tracked, projects in SLA breach, total recorded cost escalation
- **Cost by sector** across all 8 sectors · **Top states table** across 15 states
- **Progress scatter** — physical vs. financial progress, exposing reporting divergence visually
- **Cost overrun histogram** — the shape of escalation across the whole portfolio
- **SLA rule bars** — how many projects breach each of the five rules
- **Critical watchlist** — the projects that need attention this week
- **Recent analyses** — the last saved document-driven reports

Project intelligence report — the per-project deep view, organised as numbered sections: risk prediction with SHAP factors, historical precedent, GIS screening, recommended interventions (section 04), and the scenario simulator (section 05). Exportable as a brief.

Supporting surfaces: searchable and paginated project registry · GIS collision checker with an interactive map and boundary legend · document analyser · saved-analyses manager with delete confirmation · SLA breach list · secure workspace sign-in.

Scenario lab **PARTIAL** — four intervention sliders (land acquisition resolved %, funding availability %, delayed milestone count, resource allocation %) producing a live current-trajectory vs. intervention-scenario comparison on risk score, predicted delay months and cost impact. It is clearly labelled "**MODEL ESTIMATES ONLY**" in the interface because it currently runs a transparent sensitivity heuristic rather than a counterfactual model. See *Section 14, item 1*.

UX principles worth one line on the slide: empty states are always real explanations with a next action, never fake placeholder rows. Loading skeletons are sized to the content so nothing shifts. Every failure surfaces as a typed error with a retry — no response value is ever synthesised on the client to cover a backend that is down.

16 Slide 16 — The AI/ML Stack, and Why Each Choice

SLIDE 16

Four techniques, each chosen for a reason we can defend

Technique	Used for	Why this and not something else
XGBoost (gradient-boosted trees)	Delay risk classification	Best-in-class on tabular data with mixed types and non-linear interactions at this data scale. Trains in seconds, serves in milliseconds, and — critically — admits exact SHAP attribution. A neural network would give us marginally different accuracy and no defensible explanation.
SHAP (TreeExplainer)	Per-prediction explanation	Game-theoretically exact attribution for tree ensembles. Not a post-hoc approximation like LIME; the contributions provably sum to the prediction. This is what converts a score into evidence.
k-Nearest Neighbours (weighted, scaled)	Historical precedent retrieval	The required output is the actual comparable projects with their real outcomes — something an officer can independently verify. A learned embedding would be more elegant and completely unverifiable.
Deterministic weighted scoring + rule engine	Priority ranking and interventions	A conscious decision <i>against</i> ML. The layer that decides where ministerial attention goes must be inspectable line by line and tunable by policy rather than by retraining. Using ML here would have been technically showier and institutionally useless.

Supporting ML infrastructure: scikit-learn Pipelines (preprocessing and model travel together, so training and serving can never drift apart) · ColumnTransformer with median/most-frequent imputation and one-hot encoding · StandardScaler for the metric space of the similarity engine · joblib persistence with models trained offline and loaded once at startup · a saved metrics artefact so the reported performance is always the measured performance of the deployed model.

The line that ties the whole deck together: *"We used machine learning exactly where it is better than a rule — predicting an outcome from many weak interacting signals — and refused to use it where a rule is better: deciding what should happen next. That restraint is the design."*

17 Slide 17 — What Works Today vs. What Comes Next

Built and demonstrable today

- Trained XGBoost delay-risk classifier with measured held-out metrics, served through a live API
- SHAP explanation of every prediction, translated to plain-language factors
- Historical similarity search over 1,500 completed projects with real outcomes and interventions
- Composite priority scoring with dual weight profiles and full component breakdown
- Validated GIS boundary layer with geometrically correct, projected-metre spatial analysis and an interactive map
- Rule-based intervention recommender with statutory precedence
- PDF document extraction with per-field confidence and human confirmation
- Five-rule SLA monitoring with severity classification across the portfolio
- Full portfolio dashboard, project registry, GIS checker, document analyser, saved analyses, SLA view, workspace sign-in
- Bearer-token authentication with role-based access control on every endpoint

Next, in priority order

1. Counterfactual scenario modelling to replace the sensitivity heuristic
2. Live SMS/email escalation switched on, with an audit trail
3. Full national protected-area boundary ingestion and PostGIS backend
4. Integration with live government project-monitoring data sources
5. Model retraining on real outcome data, with recall as the target metric
6. OCR for scanned documents, and multilingual support
7. Institutional SSO, per-ministry role scoping, and a mobile field-reporting surface

"We are showing you a working system with known edges, not a prototype with hidden ones. Every item on the right-hand list already has its seam built into the code."

18 Section 14 — Honest Gaps & Roadmap (backup slide + Q&A armour)

Keep this as a **backup slide**, not a main one. If a judge asks "what doesn't work yet?", you open it and answer completely. Volunteering limitations with a concrete plan for each reads as engineering maturity; being caught out by one does the opposite.

GAP 1 PARTIAL

Scenario simulator is a sensitivity heuristic, not a counterfactual model

What exists: four intervention sliders producing a live comparison of current trajectory vs. intervention scenario across risk score, predicted delay months and estimated cost impact. The interface labels it **"MODEL ESTIMATES ONLY"**.

What is limited: the projected values come from a transparent, monotonic sensitivity formula anchored to the real base risk score — not from re-running the trained model on modified inputs, and not from a causal model. It shows the *direction and rough magnitude* of an intervention's effect correctly, but the specific numbers are illustrative.

The plan: feed the modified inputs back through the actual XGBoost pipeline to produce a genuine counterfactual probability, then ground the delay and cost deltas in the observed outcomes of historical projects that received that same intervention — the corpus already records `intervention_taken` and `intervention_outcome` for all 1,500 projects, so the evidence for this is already in place.

GAP 2 PARTIAL

SMS escalation is composed and previewed, never dispatched

What exists: a full MSG91 flow-API integration with template, sender and recipient configuration, recipient masking in logs, thread-safe dispatch, and a complete alert preview surfaced in the SLA panel — showing exactly what would be sent, to which role.

What is limited: delivery is switched off by configuration in this environment. The interface states this explicitly: *"NOT DISPATCHED · SMS delivery disabled"*. No live message has been sent.

Why it is off: deliberately. Sending real SMS to real numbers from a hackathon environment is the wrong default, and a fake "message sent" toast would be exactly the kind of dishonesty this system is built to avoid.

The plan: enable on deployment with a verified sender ID and an approved DLT template, add email and in-app channels, and record every dispatch in an escalation audit trail so a breach and its escalation are permanently linked.

GAP 3 PARTIAL

Boundary dataset is a 6-region demonstration set

What exists: a complete, validated boundary layer — six boundaries covering one of each major protected category (national park, wildlife sanctuary, tiger reserve, forest, Ramsar wetland, eco-sensitive zone), enough to exercise every collision class and every code path.

What is limited: it is not national coverage. A real project in an unmapped region will screen as CLEAR simply because no boundary is loaded there.

The plan: ingest the official national protected-area and forest datasets through the importer that already exists. Deterministic UUIDv5 identity means re-import is idempotent, and the validator already rejects malformed geometry with a reason — so this is a data operation, not a development effort.

GAP 4 ROADMAP

PostGIS production backend

What exists: the GIS layer is written against a storage interface with two implementations planned from the start; the GeoJSON file store is in use and is the faster path for a dataset of this size.

What is limited: the PostGIS backend with GiST spatial indexing is not the active path, so the system has not yet been demonstrated at national-dataset scale.

The plan: switch the backend once the full boundary dataset lands. No route handler, service or frontend component changes — the seam was built for exactly this.

GAP 5 PARTIAL

Data is a realistic generated corpus, not a live government feed

What exists: 5,000 current projects across 8 sectors and 15 states with labelled delay outcomes, and 1,500 completed historical projects with real-shaped outcome fields — generated by documented, validated scripts with realistic cost ranges, class balance ($\approx 40\%$ delayed) and a delay-cause distribution that matches the known pattern (land acquisition dominant, then milestone slippage, then clearances and funding).

What is limited: it is synthetic. The model's 0.75 ROC-AUC is measured on this corpus, and real-world performance will differ.

The plan: ingest live project records from the national project-monitoring systems. The entire pipeline is schema-driven with strict validation, so the integration point is a data adapter. Retraining on real outcomes is expected to change the numbers — and we would rather report a real number that moves than a synthetic one that looks better.

GAP 6 PARTIAL

Model performance has clear headroom, especially recall

What exists: measured held-out performance — accuracy 0.697, precision 0.650, **recall 0.535**, F1 0.587, ROC-AUC 0.753, on a 1,000-project held-out test set.

What is limited: recall of 0.535 means roughly half of the projects that eventually slipped were not flagged. For a triage tool, **recall is the metric that matters most** — a missed at-risk project is a worse failure than a false alarm that costs one review meeting.

The plan: threshold tuning to trade precision for recall at the operating point an officer actually wants; class weighting and calibration; richer temporal features (progress velocity, rate of milestone slippage rather than the static count); and retraining on real outcomes. All of these are ordinary, well-understood next steps — the pipeline and metrics artefact are already in place to measure whether each one helps.

GAP 7 PARTIAL

Document extraction: no OCR, no scanned-PDF support

What exists: deterministic label-based extraction from the text and table layers of digital PDFs, with per-field confidence, source notes and unit normalisation.

What is limited: a scanned image PDF has no text layer and yields nothing. The system says so explicitly rather than failing quietly — but it cannot read it.

The plan: add an OCR pre-processing stage for scanned documents, and extend label matching to Hindi and regional-language report formats. Confidence tagging and mandatory human confirmation stay in place regardless of the extraction route, so a lower-confidence OCR path never bypasses review.

GAP 8 PARTIAL

Authentication is demonstration-grade at the workspace layer

What exists on the backend: real bearer-token authentication against a SHA-256 hashed token registry loaded once at startup, with role-based access control (admin / analyst / viewer) where routes declare the permission they require rather than checking raw tokens, plus a restricted CORS policy.

What is limited: the workspace sign-in screen uses demonstration credentials for the presentation environment, and a development mode exists that bypasses token checks locally.

The plan: the authentication layer was written as a deliberate seam — it can be swapped to OIDC/JWT and institutional SSO **without modifying a single route handler**. Per-ministry and per-state role scoping, and an access audit log, follow from the same layer.

Deferred by choice — not attempted, and here is why

Capability	Position
Mobile field-reporting app	The web application is fully responsive; a native field-reporting surface for site engineers is a genuine next phase, not a wrapper around this UI.
Satellite imagery progress verification	High value and high cost. It would independently verify self-reported physical progress — the single weakest input in the whole system. Deliberately scoped out of phase one.
Multilingual interface	English-only today. Required before any state-level field deployment.
Contractor performance history	Contractor issues are currently a boolean flag. A cross-project contractor track record would be a strong predictive signal and needs a data source we do not yet have.
LLM-generated narrative summaries	Consciously excluded. Every output in this system is traceable to a record, a rule or a model contribution. We were not willing to trade that for prose.

19 Slide 18 — Impact, Feasibility & Close

SLIDE 18

Why this is adoptable, not just buildable

Impact — for the ministry

- Attention goes to the right projects, ranked and justified
- Delay risk surfaces while the window to act is still open
- Environmental conflicts found at siting, not at clearance
- Every escalation carries its evidence — file-note ready

Impact — for the citizen

- Projects finish closer to their promised date and cost
- Public money is redirected from escalation to delivery
- Decisions become inspectable rather than opaque

Feasibility

- Uses only data the monitoring system already collects
- No new field-reporting burden on any officer
- Trains in seconds; serves in milliseconds; runs on modest hardware
- Every threshold tunable by policy without retraining

Risk management

- Degrades gracefully and reports what is unavailable
- Never synthesises a value to cover a failure
- Human confirmation before any extracted value is used
- Measured facts and model estimates never presented as one

Closing line

"PRAGUKTI does not ask an officer to trust a number. It hands them a ranked list, the reasons behind each ranking, forty past projects that looked the same, what was done about them, and what worked. That is not a dashboard. That is a decision, arriving early enough to matter."

20 Q&A Preparation — the six questions you will be asked

1. "Your accuracy is only 70% — why should anyone trust it?"

Because it is a triage tool, not a verdict. ROC-AUC 0.753 means it correctly ranks a delayed project above an on-track one three times in four, which is exactly what is needed to decide review order across 5,000 projects. It never acts alone — every prediction carries its top five factors, and an officer can overrule it with one glance at the reasons. We also report the real held-out number rather than a flattering one, and we know recall is the metric to improve first.

2. "This is synthetic data. Does any of it transfer?"

The corpus is generated but validated against the known real-world pattern — realistic cost ranges, ~40% delay rate, and a delay-cause distribution dominated by land acquisition, then milestone slippage, then clearances and funding. What transfers is the entire pipeline: schema, feature engineering, training, explanation, serving. Swapping in real records is a data-adaptor change, and the metrics artefact means we will know immediately how performance moves.

3. "Why not use an LLM / deep learning?"

On tabular data at this scale, gradient boosting outperforms deep learning and gives exact SHAP attribution, which a neural network cannot. And for the layer that decides where ministerial attention goes, we deliberately chose deterministic rules over any model — that ranking must be inspectable and policy-tunable, not retrained. We also refused to put a language model on document extraction, because a hallucinated cost figure entering a risk calculation is far worse than an empty field.

4. "What is actually not finished?"

Open the backup slide and walk through all nine gaps. Lead with: the scenario simulator is a labelled heuristic, SMS is composed but not dispatched, and the boundary dataset is a six-region demo set. Then note that every one of them has its seam already in the code.

5. "How does this integrate with existing government systems?"

It consumes fields the monitoring system already collects — no new reporting burden. The API is a standard REST surface with token auth and RBAC, and the auth layer is a deliberate seam that swaps to OIDC/SSO without touching a single route handler. The GIS layer already has a PostGIS path for departmental deployment.

6. "What makes this different from a BI dashboard?"

A dashboard reports the past. This predicts, explains the prediction, proves it with historical precedent, screens the geography, ranks the project against every other project, and tells you which action to take first — with statutory obligations ordered above managerial ones. The dashboard is one screen of ten in this system, and it is the least intelligent one.

- **Colour:** deep navy (#16233d) as the base, one blue accent (#2563eb), and status colours used sparingly — green for working, amber for partial, red only for risk figures. Do not introduce a fourth hue.
- **Six words per bullet on screen.** Everything longer in this document is speaker script, not slide text.
- **One visual per feature slide.** The highest-value visuals, in order: the SHAP factor bar chart (slide 7), the GIS map with buffer ring and colliding boundary (slide 11), the priority weight breakdown (slide 9), and the architecture flow (slide 5).
- **Status tags on every feature slide.** Putting a small WORKING / PARTIAL chip on each feature slide, before anyone asks, is the single strongest credibility move available in this deck.
- **Show real screenshots, not mockups.** The dashboard, GIS checker, intelligence report and document analyser all render real data — use them.
- **Demo order if you get live time:** dashboard → open a critical project → run analysis → SHAP factors → historical matches → GIS check with a colliding coordinate → interventions. That sequence tells the whole story in about ninety seconds.